

SIPHEPHELO SIBANYONI

SOFTWARE DEVELOPER

Full-Stack Developer | Angular & React | .NET & Java/Spring Boot | Agentic AI

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PROFESSIONAL SUMMARY

Full-stack Software Developer with 3 years of experience building production web applications across .NET 8/C# and Java 17/21 (Spring Boot), with Angular (12–21, NgRx) and React front ends, over PostgreSQL/Oracle data stores, RabbitMQ/Kafka microservices, and Azure DevOps CI/CD. Resolved production performance incidents through query refactoring and caching, and led SonarQube-driven code-quality initiatives across several codebases. Recently shipped agentic AI features — human-in-the-loop workflows on .NET 8/Angular 20 with Temporal orchestration and AWS Bedrock, plus a vendor-agnostic Ollama/OpenAI LLM integration — and is looking to bring full-stack and applied-AI engineering skills to a growing development team.

PROFESSIONAL EXPERIENCE

Software Developer | Adapt IT | July 2024 – Present

- Build and maintain enterprise web applications for tertiary-education clients using Angular (12–21), TypeScript, C#/.NET 8, Java/Spring Boot, and PostgreSQL/Oracle across multiple microservices.
- Design and implement CQRS (MediatR) APIs and RabbitMQ/Kafka-based microservices following Clean Architecture principles.
- Resolve production performance incidents, including API gateway timeouts, through query refactoring and caching strategies; lead SonarQube-driven refactoring to reduce cognitive complexity across codebases.
- Build agentic AI features — human-in-the-loop workflows and LLM provider integrations (AWS Bedrock, Ollama, OpenAI) — across .NET, Java, Angular and React stacks.
- Own CI/CD release promotion across Azure DevOps GitFlow pipelines (Dev → UAT → institution TEST/PROD) for multiple tenant institutions.
- Author unit and integration tests (xUnit, Moq, Jest) to protect refactors and new features.
- Collaborate within Agile Scrum teams, conducting code reviews, debugging, and root-

cause analysis.

Graduate Software Developer | Adapt IT | July 2023 – June 2024

- Developed RESTful APIs and business services using C#, Entity Framework, and PostgreSQL.
 - Collaborated with UI/UX teams to implement design-driven frontend solutions using Angular.
 - Authored automated unit tests using xUnit and NUnit to improve software quality.
 - Participated in sprint planning, backlog refinement, estimation, and Agile ceremonies.
 - Performed application monitoring, log analysis, troubleshooting, and bug fixing.
 - Contributed to code reviews and refactoring initiatives to improve maintainability.
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KEY SKILLS

Angular (12–21) • NgRx • RxJS • React 18 • TypeScript • JavaScript • C# • .NET 8 • ASP.NET Core • Java 17/21 • Spring Boot 3 • Spring WebFlux • PostgreSQL • Oracle (PL/SQL) • SQL Server • REST APIs • CQRS/MediatR • Microservices • RabbitMQ • Apache Kafka • Docker • Azure DevOps • CI/CD • JWT/OAuth2 • BCrypt • AI/LLM Agents • AWS Bedrock • Ollama • OpenAI • Temporal • SonarQube • Unit & Integration Testing • Agile/Scrum • Clean Architecture • SOLID Principles

TECHNICAL SKILLS

Programming Languages: C#, Java, TypeScript, JavaScript, Python, SQL

Frontend Frameworks & Libraries: Angular 12–21, NgRx, RxJS, Angular Material, React 18 (Vite), Angular SSR/PWA

Backend Frameworks & Libraries: ASP.NET Core (.NET 8), Entity Framework Core, MediatR/CQRS, Dapper/Npgsql, Spring Boot 3, Spring WebFlux, MapStruct

Databases: PostgreSQL, Oracle (PL/SQL), SQL Server, SQLite, RavenDB, Liquibase, EF Core Migrations

Messaging & Async: RabbitMQ, Apache Kafka, SignalR/WebSockets, Hangfire, Quartz, Temporal (durable workflow orchestration)

AI / LLM Engineering: AWS Bedrock, Ollama (local LLM), OpenAI, Prompt Engineering, agentic workflows with human-in-the-loop approval gates, AI provider factory/strategy abstraction, OCR (Tesseract, Microblink, Azure OCR, OCR.Space)

Web Technologies: HTML5, CSS3, REST APIs, JSON, OpenAPI/Swagger

Security: JWT, OAuth2, BCrypt password/PIN hashing, PII masking, Role-Based Access Control

Testing: xUnit, NUnit, Moq, Jest, Jasmine/Karma, Selenium, integration testing (Postgres fixtures), E2E testing

DevOps & Tools: Git, Azure DevOps (CI/CD, Repos, Boards), Docker, SonarQube, Postman, Visual Studio, Visual Studio Code, PgAdmin, SQL Server Management Studio, Supabase, Kiro, Claude Code, AmazonQ

Software Engineering Practices: OOP, SOLID Principles, Clean Code, Clean Architecture, CQRS, Design-Driven Development, AI-Driven Development, Debugging, Agile/Scrum

SIGNIFICANT PROJECTS

Education – Student Applications System

Next-generation student applications platform (ITS5) for South African tertiary institutions, covering the full applicant journey — registration, document upload, APS calculation, payments and inline submission into the ITS system of record — built with .NET 8 (Clean Architecture, CQRS/MediatR) and Angular 12, with RabbitMQ-based microservices.

- Designed and shipped application analytics/reporting endpoints (completion, missing-document and payment metrics) and resolved a production gateway-timeout incident by refactoring EF Core queries and introducing in-memory response caching, promoted across multi-tenant DEV→UAT→PROD environments.
- Led a SonarQube-driven refactoring effort, reducing cognitive complexity across dozens of queries and commands and splitting controllers via abstract base classes, backed by xUnit/Moq unit tests.
- Built an Angular local-storage caching layer for lists-of-values and owned CI/CD GitFlow release promotion across multiple institutional environments in Azure DevOps.

Education – Student Registrations System

Self-service student registration platform for tertiary institutions and the front end to AdaptIT's ITS system — covering qualification/subject selection, credit transfers &

exemptions, document upload and inline submission into ITS — built with Angular 18 (NgRx) and a .NET/MediatR CQRS API over PostgreSQL.

- Delivered the full-stack Credit Transfers & Exemptions feature end-to-end, submitting prior-study credits inline to the ITS system via typed HTTP clients.
- Built and refactored responsive Angular Material/Bootstrap UI with NgRx state, SweetAlert2 notifications, and form/date validation across desktop and mobile.
- Drove a SonarQube-backed testing initiative, authoring unit tests across personal-details, subjects, qualifications and relationships features.

Education – Alumni

Alumni engagement platform for tertiary institutions — content publishing, a moderated community feed, an alumni talent pool and employment messaging — built as a JHipster application (Angular 19 + Spring Boot 3/Java 17) over Oracle/PL/SQL.

- Top contributor across both repos (96 of 233 commits on the active development repo; initialized and drove the promoted production repo), delivering across the frontend and backend.
- Built the Content Hub blog subsystem end-to-end — CRUD REST controllers, a rich-text editor, scheduled publishing and category management — plus a moderated community feed and an alumni Talent Pool with filterable, paginated profiles.
- Implemented real-time employment-inbox messaging over WebSocket and an alumni terms-and-conditions/global-settings admin flow, secured with JWT OAuth2.

Education – TVET Student Success Tracker

AI-assisted student-success and Monitoring & Evaluation platform for TVET colleges, tracking pass/retention/progression/achievement rates and surfacing at-risk students, built with Angular 17 and Spring Boot 3/Java 21 over PostgreSQL.

- One of the top two contributors overall and the largest contributor to the web app, delivering 20+ merged pull requests across frontend and backend.
- Designed and built the analytics/M&E dashboard suite (pass, retention, progression and achievement-rate views) with dynamic filters and CSV export.
- Introduced a vendor-agnostic LLM provider layer (interface + factory pattern) enabling the platform to run against OpenAI or a local Ollama model, and migrated the backend to Java 21.

Education – Student Success Agent

Agentic student early-warning platform: a rules-based risk-scoring agent (GPA + attendance) paired with an AI agent that proposes interventions through human-in-the-loop approval and escalation gates, with full provenance and an audit trail — built with .NET 8,

Angular 20, Temporal durable workflow orchestration and AWS Bedrock.

- #1 contributor on the project; built the Risk Queue digital-twin slide-over workspace, combining a risk-trajectory sparkline with a 4-driver provenance breakdown and a case workspace for agent proposals and human approval gates.
- Designed the Cases v4 UI, presenting each intervention as a closed-loop "story card" (agent reasoning, gates, action items, timeline, outcome).
- Built on a rules-based flow with an LLM-ready seam (interface-abstracted Bedrock composers), PII masking, and a full xUnit unit/integration/E2E test suite in a Dockerised one-command stack.

ITS Ignite

AI-powered idea and innovation-management platform for AdaptIT's ITS product suite, with AI duplicate detection, idea grouping, scoring, chat and OCR document ingestion, built with a React 18 (Vite) frontend and a .NET 8 clean-architecture backend over PostgreSQL.

- Established the backend foundation — the layered API/Core/Context/Domain solution, CQRS/MediatR, EF Core + PostgreSQL and the initial migration pipeline.
- Implemented authentication and role-based access control (JWT), including the signup flow and role-driven UI gating for standard vs. admin users.
- Built the user profile module and real-time notifications feed integrated with the AI idea-grouping workflow, and owned release promotion from develop to main.

EDUCATION

Bachelor of Computer and Information Sciences in Application Development
Varsity College, Sandton | 2021

Diploma in Information Technology (Software Development)
Varsity College, Pretoria | 2019

PROFESSIONAL DEVELOPMENT

- ASP.NET Core – Solid and Clean Architecture
- Azure DevOps Boards for Project Managers, Analysts and Developers
- Git and Visual Studio with Azure DevOps Repos for Developers
- OpenAPI: Beginner to Guru
- Selenium WebDriver with C# from Scratch – NUnit Framework

- SQL & PostgreSQL for Beginners
- Understanding APIs and RESTful APIs

REFERENCES

Available upon request